

Course Objective:

This course aims to provide students with a good understanding of the concepts and methods used in Linear Algebra. By the end of this course, students should develop the ability to solve problems using the methods learned.

Learning Outcomes:

- 1) Students will learn to Set up and solve systems of linear equations
- 2) Perform matrix operations as appropriate
- 3) Evaluate determinants and use their properties
- 4) Understand and use linear transformations
- 5) Work in real vector spaces
- 6) Use the concepts of basis, dimension, row rank, column rank, and nullity
- 7) Eigen values and eigen vectors, Diagonalizations
- 8) Gram Schmidt process; QR-decomposition
- 9) Fractals, Games of strategy
- 10) Graph theory, cryptography, linear transformations.

Reference Books:

1. Elementary Linear Algebra (Application Version) by Howard Anton, charis rorres, anton kaul 12th Edition (Australia and new Zealand).
2. Elementary Linear Algebra by Howard Anton 10th Edition.
3. A first course in linear algebra, RA Beezer, <http://linear.ups.edu/>
4. Linear Algebra and its Applications by Gibert Strang.

Course Outline and Schedule

Week (Mon-Sat)	Topics	Text Book	Tests & Assignments
<u>Wk-1</u> 10-Feb	Introduction to systems of linear equations, Matrices, Matrix Operations and Gaussian Elimination.	Chp 1	
<u>Wk-2</u> 17 Feb	Algebraic Properties of Matrices, Elementary Matrices and a Method for Finding A-1	chp 1	
<u>Wk-3</u> 24 Feb	Determinants by Cofactor Expansion and Row reduction, Properties of determinant and Cramer's Rule	chp 2	
<u>Wk-4</u> 03-Mar	Intro to vectors, dot and cross product, distance in R^n	Chp 3	Class test 1
<u>Wk-5</u> 10-Mar	Real vector spaces Subspaces	Ex 4.1 (1-12) Ex 4.2 (1-16)	
<u>Wk-6</u> 17-Mar	Spanning sets Linear independence	Ex 4.3 (1-10) Ex 4.4 (1-14)	
<u>Wk-7</u> 24-Mar	Coordinates and Basis Dimension	Ex 4.5 (1-18) Ex 4.6 (1-6)	

<u>Wk-08</u> 31-Mar	Spring Break		Assignment 1
<u>Wk-9</u> 07-Apr	Rank, Nullity, and the fundamental Matrix space	Ex 4.9 (1,2)	Class Test 2
<u>Wk-10</u> 14 Apr	Mid Exam		
<u>Wk-11</u> 21 Apr	Eigenvalues and Eigenvectors Diagonalization	Ex 5.1 (1-12) Ex 5.2 (5-20)	
<u>Wk-12</u> 28 Apr	Dynamical Systems and Markov Chains	Ex 5.5 (1-12)	Class Test 3
<u>Wk-13</u> 05 May	Inner products, Gram-Schmidt Process	Ex 6.1 (1-26)	Assignment 2
<u>Wk-14</u> 12 May	Gram-Schmidt Process; QR-Decomposition LU-Decompositions	Ex 6.3 (27-31, 45-49) Ex 9.1	
<u>Wk-15</u> 19 May	Fractals, Games of strategy	Ex 10.11 Ex 10.6 (1-5)	Class Test 4
<u>Wk-16</u> 26 May	Graph Theory, Cryptography	Ex 10.5 (1-7), Ex 10.13 (1-6)	
<u>Wk-17</u> 02 june	General Linear Transformations Compositions and inverse Transformations	Ex 8.1 (1-22) Ex 8.2 (1-8)	
<u>Wk-18</u> 09 Jun	Preparation week		
<u>Wk-19</u> 16 Jun	Final Exams		
<u>23 Jun-11 july</u>	Semester break		
27 Jun	Posting of Result in CMS		

Division of Marks:

Class Tests (3)	20
Assignments (3)	10
Mid Term Tests (2)	30
Final Exam	40
Total	<u>100</u>

GRADING TABLE			
ABSOLUTE SCALE if no. of students in Final Exam are up to 20, otherwise RELATIVE SCALE			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A–
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B–
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C–
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F